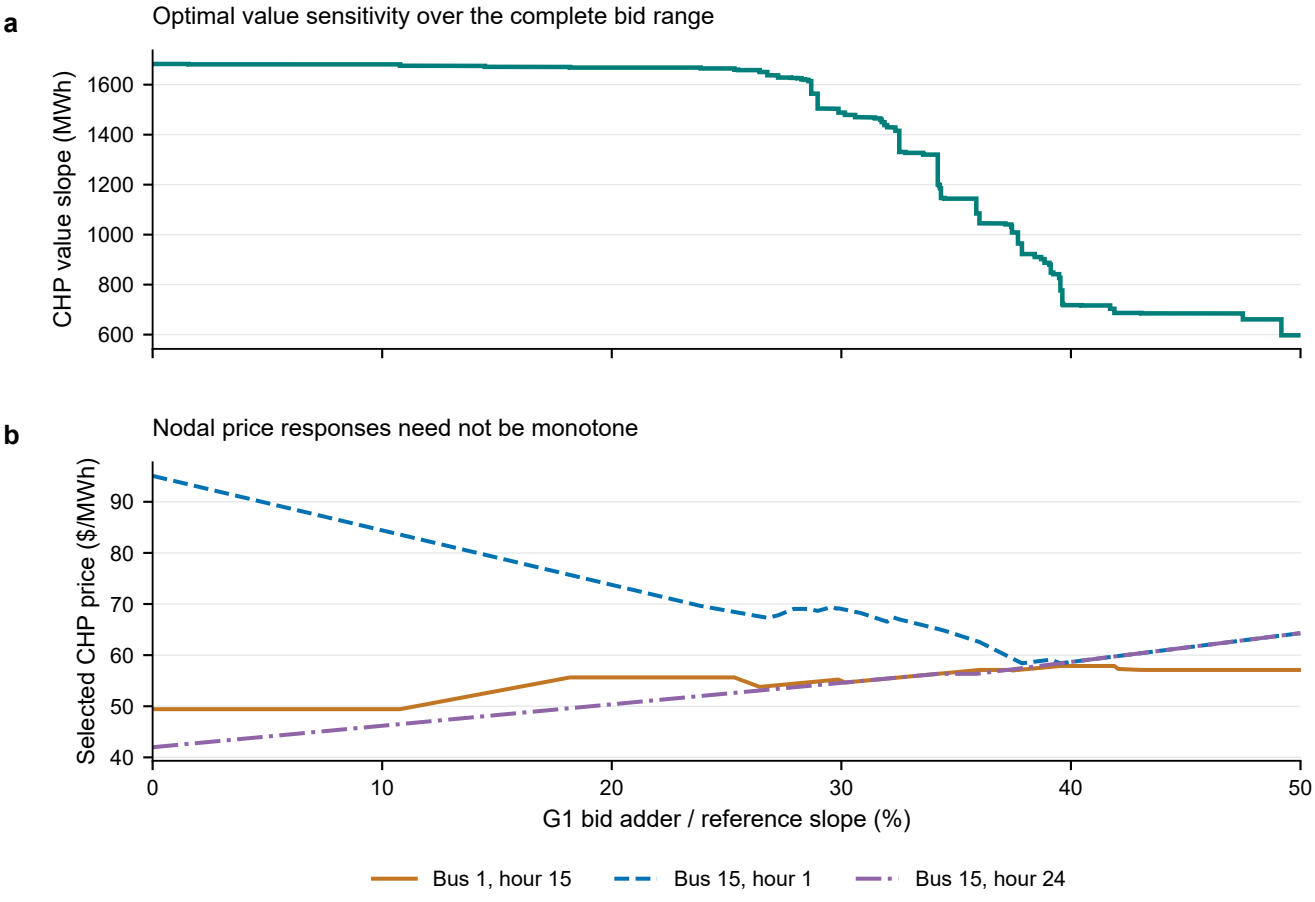
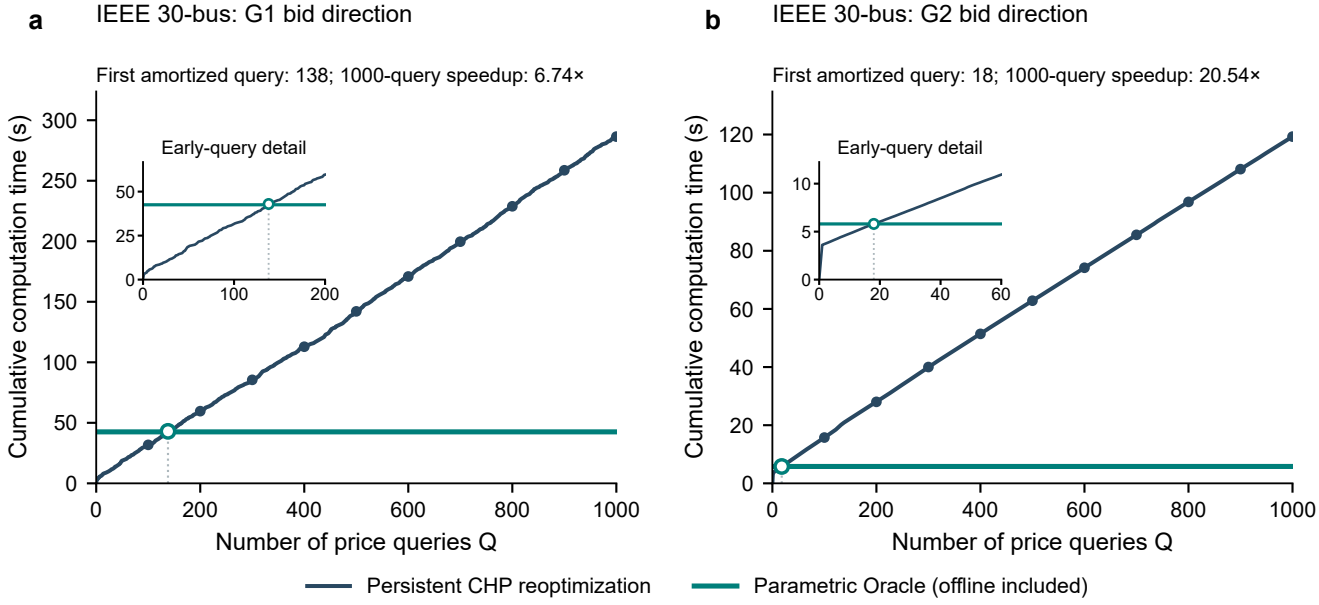
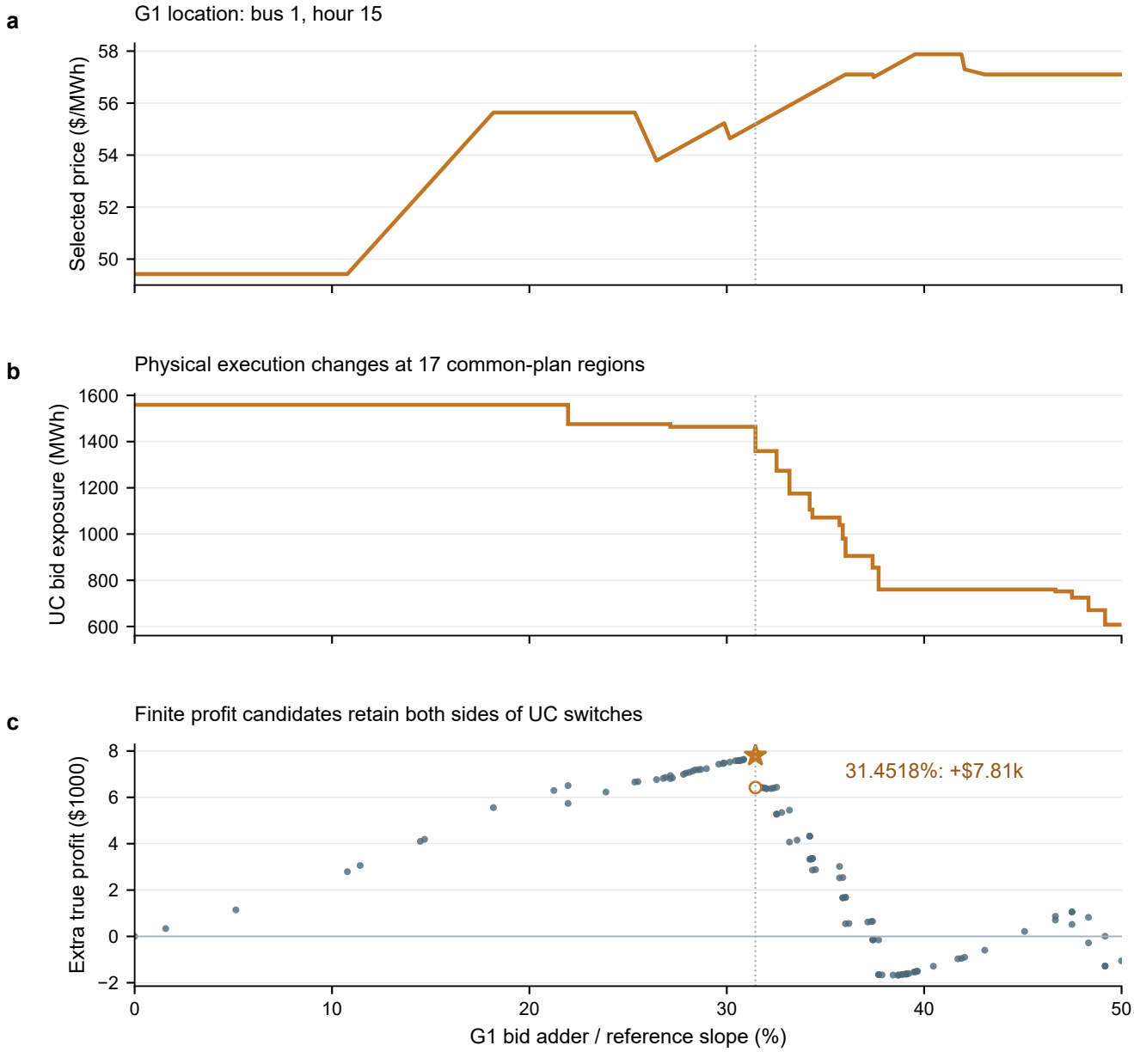
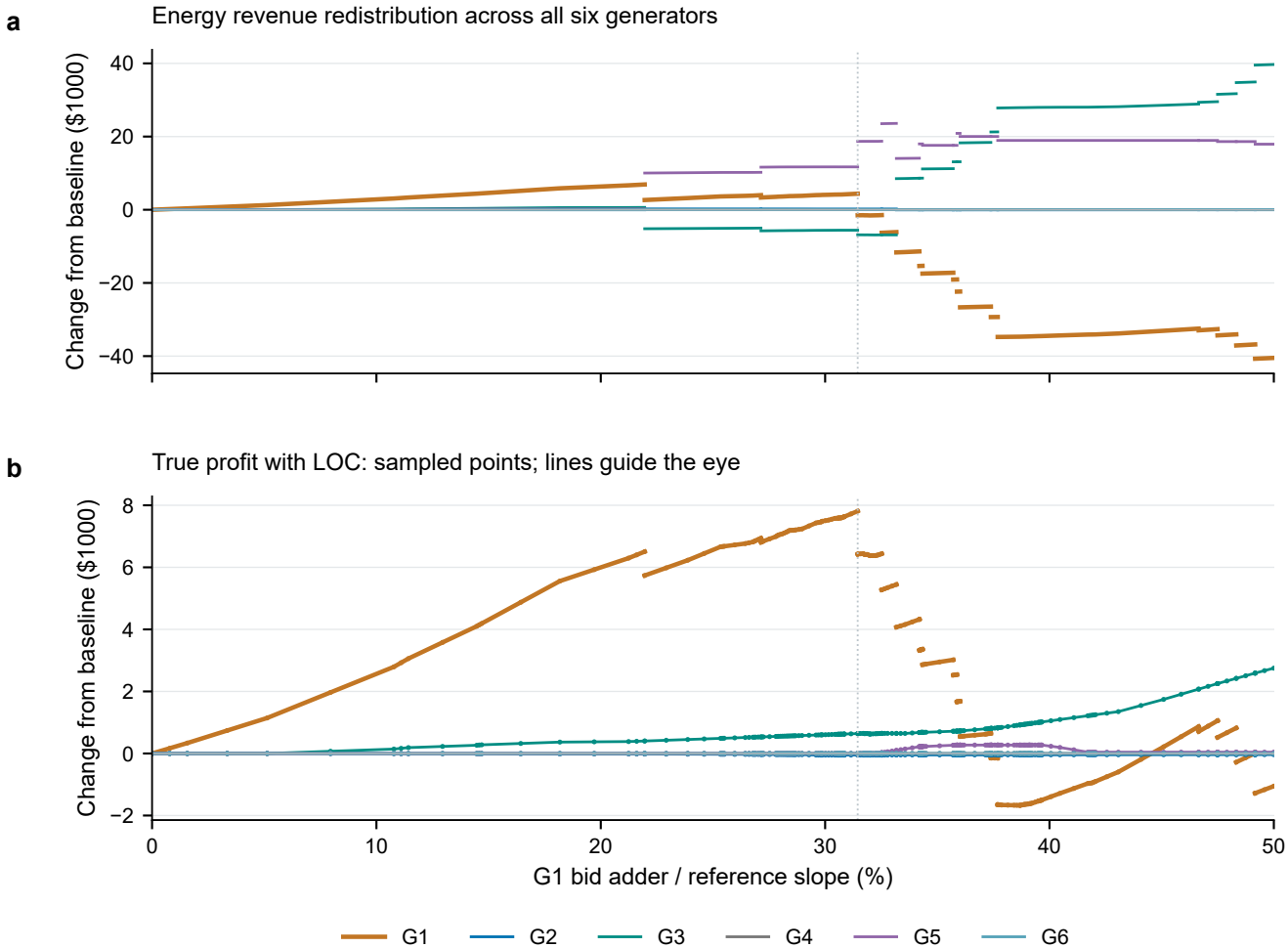
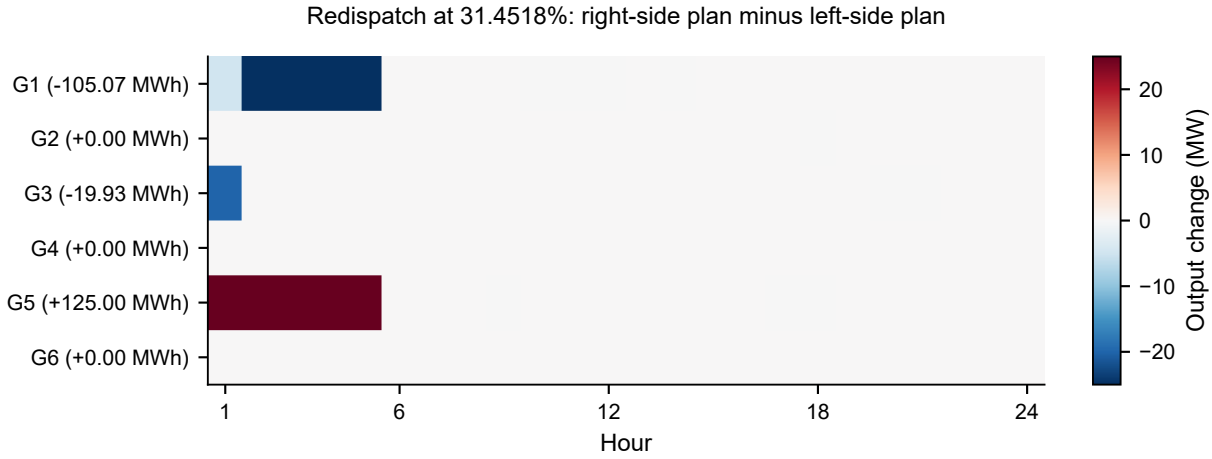

a
Optimal value sensitivity over the complete bid range
CHP value slope (MWh)
1600
1400
1200
1000
800
600
b
Nodal price responses need not be monotone
Selected CHP price ($/MWh)
90
80
70
60
50
40
0
10
20
30
40
50
G1 bid adder / reference slope (%)
Bus 1, hour 15
Bus 15, hour 1
Bus 15, hour 24


a
IEEE 30-bus: G1 bid direction
First amortized query: 138; 1000-query speedup: 6.74×
Cumulative computation time (s)
Number of price queries Q
Early-query detail
b
IEEE 30-bus: G2 bid direction
First amortized query: 18; 1000-query speedup: 20.54×
Cumulative computation time (s)
Number of price queries Q
Early-query detail
Persistent CHP reoptimization
Parametric Oracle (offline included)


a
G1 location: bus 1, hour 15
Selected price ($/MWh)
58
56
54
52
50
b
Physical execution changes at 17 common-plan regions
UC bid exposure (MWh)
1600
1400
1200
1000
800
600
c
Finite profit candidates retain both sides of UC switches
Extra true profit ($1000)
8
6
4
2
0
−2
31.4518%: +$7.81k
0
10
20
30
40
50
G1 bid adder / reference slope (%)


a
Energy revenue redistribution across all six generators
Change from baseline ($1000)
40
20
0
−20
−40
b
True profit with LOC: sampled points; lines guide the eye
Change from baseline ($1000)
8
6
4
2
0
−2
0
10
20
30
40
50
G1 bid adder / reference slope (%)
G1
G2
G3
G4
G5
G6


Redispatch at 31.4518%: right-side plan minus left-side plan
G1 (-105.07 MWh)
G2 (+0.00 MWh)
G3 (-19.93 MWh)
G4 (+0.00 MWh)
G5 (+125.00 MWh)
G6 (+0.00 MWh)
1
6
12
18
24
Hour
20
10
0
−10
−20
Output change (MW)